Data model (ER)

English version of ../../architecture/er-model.md. The ER diagram follows the actual SQLAlchemy models (backend/models/entities.py). It can be exported to PNG/PDF with GitHub's renderer or the Mermaid CLI; the source is versioned with the code.

Ready artefacts sit next to this file: er.mmd — the extracted Mermaid source, er-model.pdf — the rendered diagram, er.archimate — the same schema in ArchiMate 3 for Archi. Rebuild: python scripts/extract_er_diagram.py, then mmdc -i er.mmd -o er-model.pdf.

```
erDiagram
    USERS ||--o{ INTERACTIONS : "assigned_kam"
    USERS ||--o{ ACTIONS : "author"
    USERS ||--o{ COMMENTS : "author"
    USERS ||--o{ ATTACHED_FILES : "uploaded_by"
    USERS ||--o{ ACTION_LOGS : "user"

    UNIVERSITIES ||--o{ INTERACTIONS : "university"
    IT_DIRECTIONS ||--o{ IT_PRODUCTS : "direction"
    IT_PRODUCTS ||--o{ INTERACTIONS : "product"
    WORKFLOW_STAGES ||--o{ INTERACTIONS : "stage"

    INTERACTIONS ||--o{ ACTIONS : "tasks"
    INTERACTIONS ||--o{ COMMENTS : "comments"
    INTERACTIONS ||--o{ ATTACHED_FILES : "files"

    USERS {
        int id PK
        string email UK
        string full_name
        string hashed_password
        enum role "user|manager|admin"
        bool is_admin
        bool is_active
    }

    UNIVERSITIES {
        int id PK
        string name
        string city
        string contact_person
        string contact_email
        string contact_phone
    }

    IT_DIRECTIONS {
        int id PK
        string name UK
    }

    IT_PRODUCTS {
        int id PK
        string name UK
        int direction_id FK
    }

    WORKFLOW_STAGES {
        int id PK
        string code
        enum scope "b2b|b2c"
        string name
        int order
```

```
    string color
    bool is_active
}

INTERACTIONS {
    int id PK
    int university_id FK
    int product_id FK
    int stage_id FK
    enum scope "b2b|b2c"
    string contract_number
    string contract_date
    int assigned_kam_id FK
    string university_specialist
    text notes
    bool is_active
}

ACTIONS {
    int id PK
    int interaction_id FK
    string title
    text description
    string due_date
    bool is_completed
    int author_id FK
}

COMMENTS {
    int id PK
    int interaction_id FK
    text text
    int author_id FK
}

ATTACHED_FILES {
    int id PK
    int interaction_id FK
    string filename
    string file_path
    int size
    string mime_type
    int uploaded_by FK
}

ACTION_LOGS {
    int id PK
    int user_id FK
    string action "CREATE|UPDATE|DELETE"
    string entity_type
    int entity_id
    text old_value
```

```
        text new_value
        string ip_address
    }
```

Integrity rules

Relationship	On delete	Meaning
`interactions.university_id → universities.id`	`CASCADE`	A university with interaction history is not deleted in isolation — the history goes with it
`interactions.product_id → it_products.id`	`SET NULL`	Deleting a product does not break the card
`interactions.stage_id → workflow_stages.id`	`RESTRICT`	An active stage cannot be deleted while it holds cards — move the tasks first
`interactions.assigned_kam_id → users.id`	`SET NULL`	An employee leaving does not delete the requests
`actions/ comments / attached_files .interaction_id`	`CASCADE`	Child records live only inside the card
`*_logs.user_id → users.id`	`SET NULL`	Audit records survive the deletion of a user

Indexing

Indexes cover the foreign keys and filter columns (`interactions.university_id`, `interactions.stage_id`, `action_logs.user_id`) and the frequently filtered strings (`universities.name`). This keeps board filtering and report building acceptable at the target load of 300+.

B2B and B2C workflows

The two stage sets differ by `workflow_stages.scope`: `WORKFLOW_STAGES` (the 14 stages of the IT School) for B2B and `B2C_STAGES` (a simplified funnel "request → payment → training → completed") for B2C. There is no versioning: a stage change applies to all active requests at once.